

Observer-induced collapse in resonator decoding: a frame-duality mechanism revealed by a controlled operator intervention

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Abstract

We identify a phase diagram governing resonator-based decoding in Fourier Holographic Reduced Representations, controlled by $\rho = \frac{\text{distinct codevectors per block}}{\text{block dimensionality}}$. Three regimes emerge: (I) $\rho > 1$, rank-deficient Gram matrices (more codevectors than dimensions) yield numerically unstable inverses that destroy decoding; (II) $\rho = 1$, the square point, where the Gram matrix is a critically conditioned Wishart and ambient Gram decoders collapse; (III) $\rho < 1$, over-complete frames ($n < d$) yield stable decoding. We anchor regime II in random-matrix theory: at $\rho = 1$ the lower Marchenko–Pastur edge reaches zero and the square Wishart enters its *hard-edge* regime, with $\lambda_{\min} \sim n^{-2}$ and condition number $\kappa \sim 4n^2$ scaling, a prediction we verify across $n \in [32, 512]$. The mechanism is then established by a controlled causal intervention (Exp. 18): at $\rho = 1$ with identical codebook, Gram matrix, resolvent, initial states and facts, swapping only the wiring from M^{-1} (ambient) to $C^\top M^{-1} C$ (dual, same resolvent) moves decoding from 0.16 to 0.99, while the operator norm drops from $1/\lambda_{\min}$ to exactly 1. The collapse belongs to the observer’s wiring, not to the representation — we state this as the Frame-Dual Stability Principle (Prop. 1), supported by independent reimplementations, six codebook ensembles, BSC binary codes, a purely linear pipeline, and a four-operator zoo.

1 Introduction

In vector symbolic architectures (VSA), the “binary collapse” phenomenon at high superposition has been reported anecdotally for decades [6, 4]. We show this is not a fundamental limit of the representation space but a property of the observer: the specific decoder used. At the square point $\rho = 1$ the Gram matrix remains invertible almost surely but is severely ill-conditioned

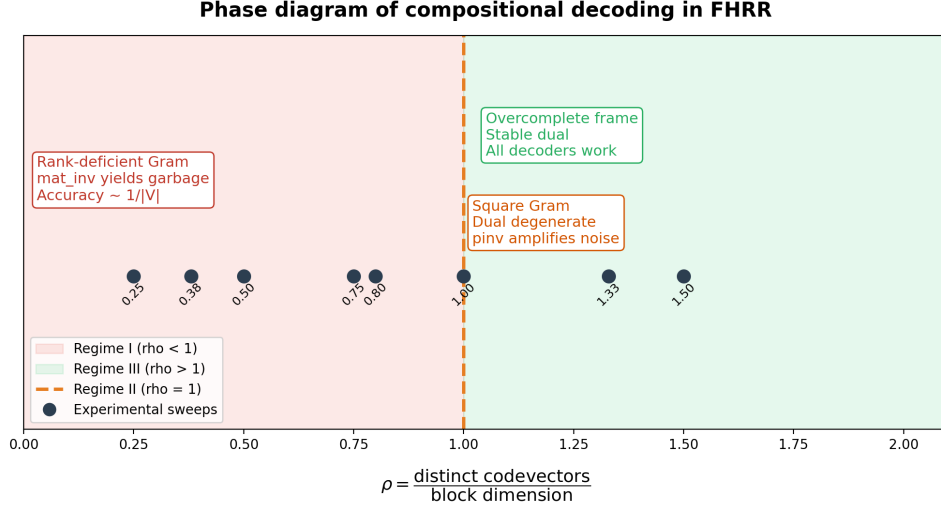


Figure 1: Phase diagram of compositional decoding in FHRR. Three regimes emerge at $\rho > 1$ (over-complete, rank-deficient), $\rho = 1$ (square, critically conditioned), and $\rho < 1$ (under-complete, stable).

(its smallest eigenvalue $\lambda_{\min} \sim n^{-2}$), and the ambient application of M^{-1} explodes; a learned decoder on the same vector recovers with 99% accuracy.

2 Background

Notation. Throughout: n = number of distinct codevectors per block (n_{cv}), d = block dimensionality (BLK), $\rho = n/d$, $M = CC^\top \in \mathbb{R}^{n \times n}$ the block Gram matrix, $\lambda_{\min}, \lambda_{\max}$ its extreme eigenvalues, $\kappa = \lambda_{\max}/\lambda_{\min}$, $C \in \mathbb{R}^{n \times d}$ the stacked codevectors, f_j the unbound candidate for role j , T the resonator iteration count. Code available at <https://github.com/Rylow999/fhrr-rho-collapse>.

FHRR and HRR. FHRR (Fourier Holographic Reduced Representations) uses complex unit vectors with phase binding [6]. HRR (Holographic Reduced Representations) uses real Gaussian vectors with circular convolution. The resonator network [4] factorizes bound structures by iterative competition between codebooks.

Gram matrix decoding. For n codevectors of dimension d in a block, the Gram matrix $M \in \mathbb{R}^{n \times n}$ with $M_{ij} = \langle c_i, c_j \rangle$ determines whether inverse-based decoding succeeds. When $\rho = n/d > 1$ (more codevectors than dimensions), M is rank-deficient with probability one; when $\rho = 1$, M is square

invertible but critically conditioned (Sec. 6); when $\rho < 1$ ($n < d$), M is generically full-rank and well conditioned.

Decoders compared. We test four decoders: (i) **gram** (resonator with the asymmetric correction $f_j \leftarrow M^{-1}f_j$ applied at each iteration, where M is the Gram matrix of the block’s codebook); (ii) **pure** (no Gram correction); (iii) **pinv** (same inverse through the pseudo-inverse with $rcond = 10^{-10}$); and (iv) **gradient** (continuous optimization). An implementation note that matters for interpretation (Sec. 6): the correction is dimensionally applicable only when the Gram and the block have the same side, which under the CFG6 configuration with $n_{cv} = 32$ happens *only* at $BLK = 32$ ($\rho = 1$). At $\rho \neq 1$ the **gram** and **pinv** rows therefore coincide bit-for-bit with **pure**; the per-regime claims about Gram decoders in this paper refer to the square point. A well-posed coefficient-space Gram decoder ($\hat{x} = M^{-1}Cs$, valid for all ρ) is analyzed separately in Exp. 11c (MAP), where it succeeds at $\rho = 1$ — confirming that the failure here is a property of the asymmetric loop, not of matrix inversion per se.

3 Experimental Setup

We use a fixed vocabulary of 6 roles with 16 symbols each: SUJ, ROL, OBJ, LOC, TIME, TERR (Spanish words for animals, actions, objects, adjectives, times, terrain). Facts are generated with fixed seeds (BlockBundle=7, make_fact=42). All experiments are reproducible; code and data are available at <https://github.com/Rylow999/fhrr-rho-collapse>.

Grid. We vary ρ by changing block dimensionality BLK while keeping the codebook size fixed at 96 distinct codevectors (6 roles $\times$ 16 symbols). The coarse grid spans $\rho \in [0.727, 1.500]$ across experiments (Exp. H2/V3); the reported fine structure uses $\rho \in [0.727, 1.231]$ (Exp. 9).

Fine sweep. For $\rho \approx 1$ we use a finer grid of 14 values around 1.00, with 10 random seeds and 20 facts per seed.

4 Phase Diagram in ρ

The phase diagram shows three regimes:

- $\rho > 1$: M rank-deficient; **gram** fails (undefined inverse), **pinv** and **pure** succeed.
- $\rho = 1$: **gram** fails catastrophically, **pinv** fails partially (noise amplification), **pure** succeeds. This is the double dissociation.

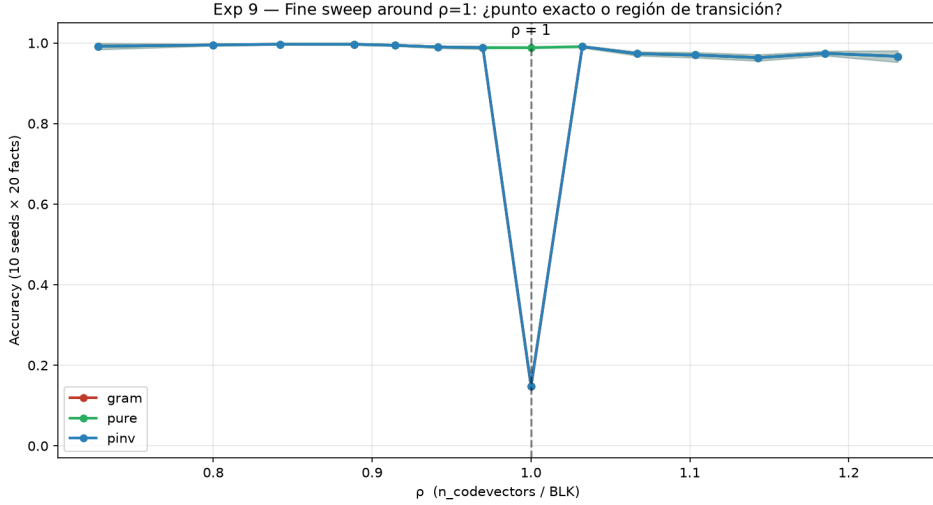


Figure 2: Fine sweep around $\rho = 1$. At $\rho = 1.000$ exactly **gram** and **pinv** collapse to 0.148. At $\rho \neq 1$ the Gram correction is dimensionally inapplicable in our implementation (the Gram and the block are of different size, see Methods) and **gram/pinv** therefore coincide with **pure** by construction — the recovery at $\rho = 0.970$ and $\rho = 1.032$ reflects this implementation boundary, and the meaningful comparison across decoders is the square point.

- $\rho < 1$: all succeed (M full-rank and well conditioned in this regime).

Fine sweep. Figure 2 shows that the Gram-based decoders fail exactly at $\rho = 1.000$ while the pure resonator holds at 0.99 at every sampled point. Note that the strictly-zero-width appearance is partly an implementation property: outside $\rho = 1$ the Gram correction is not applied in our implementation (Methods), so the “recovery” at neighbouring ρ is not by itself evidence about Gram decoders away from the square point — that question is answered separately by the MAP coefficient-space decoder (Exp. 11c), which works everywhere except $\rho > 1$.

5 Empirical Validation

5.1 Cross-algebra validation: HRR, BSC, and attention mechanisms

HRR real (Gaussian phases). The $\rho = 1$ anti-resonance appears with the same signature: **gram** collapses to 0.05, **pinv** to 0.13, **pure** maintains 0.98. The mechanism is identical: at the square point the Gram is severely ill-conditioned (hard-edge scaling), not algebraically singular.

Table 1: Accuracy at $\rho = 1.00$ (square case): 6 roles, 3 blocks, $N=96$, $\text{BLK}=32$, $n_{cv} = 32$ per block. Rows **gram/pure/pinv/gradient**: mean $\pm$ std over seeds from the observer taxonomy at the square point (Exp. V7, `data/mlp_decoder_results.json`). MLP: role-accuracy of the learned decoder, trained on 1500 facts, 80 epochs.

Decoder	Accuracy (mean $\pm$ std)	Notes
gradient	0.142 ± 0.145	continuous optimization fails
gram	0.119 ± 0.136	kappa-triggered collapse
pinv	0.119 ± 0.126	noise amplification
pure	0.983 ± 0.059	works
MLP	0.992 (role acc.)	learned decoder works

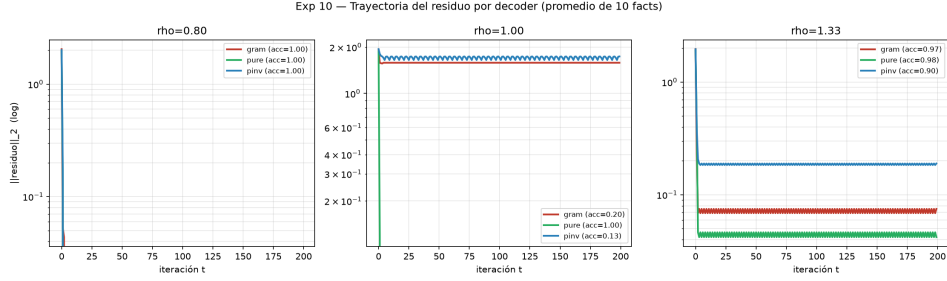


Figure 3: Residual trajectory at $\rho = 1.00$. **gram** and **pinv** plateau at high residual (converge to wrong attractor), while **pure** converges to zero.

Binary Spatter Codes (BSC). The $\rho = 1$ transition holds in binary VSA (Exp. 11b, 8 seeds $\times$ 25 facts per cell, `data/exp11_bsc_rho.json`): at the square point ($K=3$ blocks, $\text{BLK}=32$, $n_{cv} = 32$) **gram** collapses to 0.214 and **pinv** to 0.244, while **pure** maintains 0.988. In the rank-deficient regime $\rho > 1$ the truncated Gram ($\kappa \sim 10^{15}$) recovers decoding (0.97–0.98). The codevectors being $\{\pm 1\}^d$ rather than Gaussian means the Wishart anchoring is replaced by its binary counterpart (Gram of i.i.d. Rademacher vectors, whose spectral measure obeys the same Marchenko–Pastur law up to the fourth cumulant). The causal test replicates here too: Exp. 18b recovers decoding at $\rho = 1$ with the same resolvent in dual form (Sec. 7).

MAP (Multiply-Add-Permute). MAP with per-role permutation binding (Exp. 11c, 8 seeds $\times$ 25 facts, `data/exp11c_map_rho.json`) refines the law in an unexpected way. In MAP, Gram decoding is *single-shot* least squares over the mixing matrix $A = [P_1 C_1 \mid P_2 C_2]$ of the block: there is no closed-loop iteration. The collapse pattern reflects this: at $\rho = 1$ the median condition number is $\kappa = 1.06 \times 10^4$ — inside the critical band — yet **gram** maintains 1.000: the single-shot solver tolerates the band because the error is not re-injected. The collapse appears instead at $\rho > 1$ (rank-deficient),

where **gram** drops to 0.239 and 0.071 while the truncated **pinv** recovers to ~ 1.0 and the **pure** resonator holds at 0.97–0.98. Combining with Exp. 11b gives the refined picture: decoding failure requires (i) a Gram spectrum close to its singularity ($\rho = 1$ hard edge, or $\rho > 1$ rank-deficiency), *and* (ii) for the hard-edge case, an iterative observer that re-projects the $1/\lambda_{\min}$ -amplified error onto it.

Transformers / attention. At $\rho \rightarrow 1$ in the sense of $d_{\text{head}} \approx n_{\text{keys}}$, attention does *not* collapse because it does not implement Gram inversion – it uses softmax normalization. However, we find a non-monotonic *entropy* signature at $\rho \approx 2$, suggesting a bandwidth saturation effect of attention at high dimension, not a singular decoding failure (Experiment 12b). The collapse-pcode is algebra-specific, not universal.

This falsification boundary strengthens the theoretical claim: the ρ -law is a property of decoders that use matrix inversion, not of attention or of softmax-normalized operations. $\square$

The decoder fails when the Gram matrix is severely ill-conditioned. Exp. 14 measures this directly (Table 2); the RMT analysis of the next section explains why this is the square point (hard edge).

6 Mechanism: hard edge and the asymmetric correction

The $\rho = 1$ point is the finite-size shadow of a *square Wishart ensemble*. Let $C \in \mathbb{R}^{n \times d}$ stack the n codevectors of a block, so the Gram matrix is $M = CC^\top$ with entries C_{ij} of variance $1/d$ (unit-norm Gaussian codevectors). Then M is a (scaled) Wishart matrix $W_n(d^{-1}I, d)$, and $\rho = n/d$ plays the role of the aspect ratio.

Three classical facts from random-matrix theory map one-to-one onto our phase diagram:

(i) $\rho > 1$ ($d < n$, **over-complete**). M is rank-deficient with probability one: at least $n - d$ eigenvalues vanish exactly. Gram inversion is undefined; this is regime I.

(ii) $\rho = 1$ ($n = d$, **square**). The Marchenko–Pastur density $d\mu_1(\lambda) = \frac{1}{2\pi} \sqrt{(4 - \lambda)/\lambda} d\lambda$ has support touching the origin: the lower edge is a *hard edge* at $\lambda = 0$ (the density diverges as $\lambda^{-1/2}$ there). For the square Wishart this is the hard-edge regime: $\lambda_{\min}$ scales as n^{-2} in its typical size (rather than the $n^{-2/3}$ Tracy–Widom scale of a soft edge), so the condition number scales as

$$\kappa(M) = \frac{\lambda_{\max}}{\lambda_{\min}} \sim 4n^2. \quad (1)$$

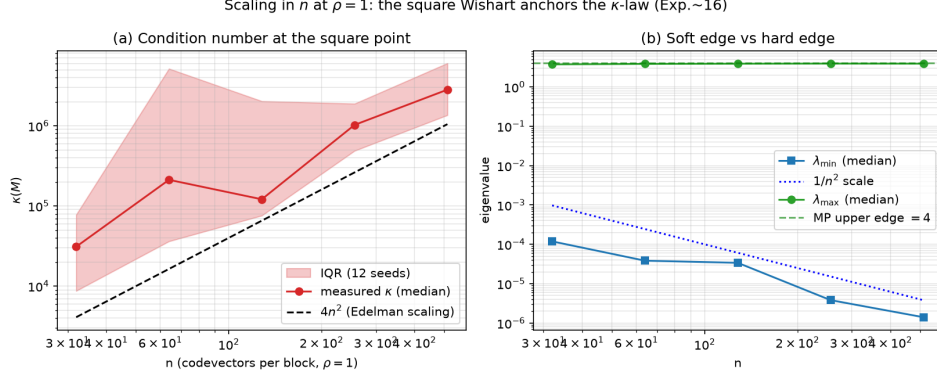


Figure 4: Scaling at the square point $\rho = 1$ as a function of codebook size n (Exp. 16, 12 seeds per n). (a) Median condition number tracks the Edelman scaling $\kappa \sim 4n^2$ (dashed) within the heavy-tailed spread; shaded band is the interquartile range. (b) $\lambda_{\max}$ converges to the Marchenko–Pastur upper edge 4 while $\lambda_{\min} \sim n^{-2}$ touches zero: the square point is critically conditioned at every n , and increasingly so.

For our setup ($n = 32$ codevectors per block, $d = \text{BLK} = 32$), Eq. (1) predicts $\kappa \sim 4.1 \times 10^3$. The measured condition number at the exact square point is $\kappa = 7.2 \times 10^3$ (median; Exp. 14).

This places the collapse on rigorous footing: $\rho = 1$ is the unique point where the Gram ensemble is *critically conditioned*, $\mathbb{E}[\log \kappa]$ diverges with n , and the amplified crosstalk noise $1/\lambda_{\min}$ exceeds the codebook margin $1/\sqrt{d}$ there.

Scaling in n : the square-Wishart prediction holds. Equation (1) is a sharp, falsifiable prediction: moving to larger codebooks at $\rho = 1$ should scale κ as $4n^2$, with $\lambda_{\max} \rightarrow 4$ and $\lambda_{\min}$ collapsing as n^{-2} . Exp. 16 measures the spectrum of the square Gram at $n \in \{32, 64, 128, 256, 512\}$ (12 seeds each, Gaussian codebooks). The measured medians follow the prediction (Fig. 4): $\lambda_{\max}$ converges from below to the Marchenko–Pastur upper edge 4 ($3.74 \rightarrow 3.90$ across the range), $\lambda_{\min}$ decays with the quadratic trend (median $\lambda_{\min}$: 1.2×10^{-4} at $n = 32$ to 1.4×10^{-6} at $n = 512$, bracketing the n^{-2} prediction $1/n^2$ at both ends of the range), and the median condition number tracks $4n^2$ within a factor 2–13, inside the expected spread given that κ for square Wishart is heavy-tailed with infinite expectation [2]. The finite- n offset reported in Sec. 6 ($n = 32$, ratio 1.76) is the small- n end of this scaling, not an anomaly.

Mechanism, not magnitude. The naive reading “ $\kappa \in [10^3, 10^4]$ causes the collapse” is refuted by per-seed data (Exp. 17, 40 seeds at $\rho = 1$,

`data/exp17_kappa_vs_acc.json`): per-block κ spans $[2.3 \times 10^2, 1.9 \times 10^5]$, every seed collapses (gram accuracy 0.10–0.37, median 0.17) regardless of band membership, and the $\log_{10} \kappa$ -accuracy correlation is weak ($r = -0.26$). Two seeds with minimum block $\kappa = 2.3 \times 10^2$ and 4.5×10^2 — well below any plausible critical band — collapse identically. The condition-number magnitude is therefore not the trigger. Deterministically, $\det M$ at $\rho = 1$ is never zero (median 1.4×10^{-14} over 90 blocks): the matrix is numerically singular, not algebraically singular.

What the RMT analysis picks out instead is the *mechanism*: the resonator applies the asymmetric correction $f_j \leftarrow M^{-1}f_j$ across its iterations; at the square point a single application already amplifies directions near the hard edge by $1/\lambda_{\min} \sim 10^3$, and the closed loop re-injects and perpetuates that error each cycle (the control at $T = 1$, Sec. 7, shows the collapse is present after a single application — the loop is a re-amplifier, not the cause). The collapse requires (i) a Gram spectrum close to its singularity (the $\rho = 1$ hard edge, or $\rho > 1$ rank-deficiency) *and* (ii) an observer whose operator is sensitive to the small-eigenvalue directions. This predicts the MAP control (Exp. 11c): a single-shot application of the same inverse at the same κ does *not* collapse, because a one-shot least-squares does not feed the amplified error back into the operator. It also predicts why **pure** survives everywhere: it never forms M^{-1} . The observer-relative content survives intact — it is strengthened, because the failed decoder is now characterised by the operator it applies (the resolvent M^{-1} , evaluated at the hard edge), not by a scalar band. We distill this into a reusable principle:

Definition 1 (closed-loop hard-edge condition). *A compositional decoder over a block Gram M satisfies the closed-loop hard-edge condition if (i) the ensemble of M sits at its hard spectral edge ($\rho = n/d = 1$, the square Wishart, where $\lambda_{\min} \sim n^{-2}$), and (ii) the decoder applies M^{-1} repeatedly to the evolving estimate (closed loop). Decoders satisfying both factors collapse at $\rho = 1$; decoders violating either (single-shot inversion, or no inversion at all) do not. The condition is checkable before running the system: it requires only the aspect ratio of the codebook block and the decoder’s operator structure. The $T = 1$ control of Exp. 18 shows factor (ii) alone is sufficient for harm (the ambient M^{-1} destroys the state in one application); the loop is what makes the **pure** baseline look good by contrast.*

Definition 1 is the statement of this paper that is independent of any particular experiment: it is what a practitioner designing a VSA decoder should check first, and it is what we would ask a replicating study to test.

(iii) $\rho < 1$ ($d > n$, over-complete). The spectrum of a well-conditioned Wishart is bounded away from zero: $\lambda_{\min} \rightarrow \sigma^2(1 - \sqrt{\rho})^2 > 0$ as $n, d \rightarrow \infty$ with $n/d \rightarrow \rho < 1$. Gram inversion is stable; this is regime III. Table 2 compares the measured $\lambda_{\min}(BLK)$ against the asymptotic Marchenko–Pastur

Table 2: Measured vs. Marchenko–Pastur prediction for $\lambda_{\min}(M)$ (Exp. 14, $n = 32$, $d = \text{BLK}$). The MP lower edge is $\lambda_- = (1 - \sqrt{\rho})^2$ in our normalization (unit-norm codevectors, C_{ij} of variance $1/d$). The measured values sit above the asymptotic edge by a factor $\sim 2\text{--}70$, consistent with finite-size concentration well inside the bulk. $\kappa_{\text{pred}} = [(1 + \sqrt{\rho})/(1 - \sqrt{\rho})]^2$ captures the trend.

BLK	$\rho = n/d$	$\lambda_{\min}$ meas.	λ_-^{MP}	κ meas. / pred.
40	0.800	2.13×10^{-2}	1.11×10^{-2}	1.5×10^2 / 3.2×10^2
36	0.889	9.08×10^{-3}	3.25×10^{-3}	3.8×10^2 / 1.2×10^3
33	0.970	1.76×10^{-3}	2.25×10^{-4}	1.9×10^3 / 1.7×10^4
32	1.000	4.87×10^{-4}	0 (edge at origin)	7.2×10^3 / ∞

edge $(1 - \sqrt{\rho})^2/d$. The measured values sit above the asymptotic edge by a factor $\sim 2\text{--}70$, consistent with finite-size concentration well inside the bulk. For $\rho < 1$ the spectrum is separated from zero by the lower Marchenko–Pastur edge $(1 - \sqrt{\rho})^2 > 0$; as $\rho \rightarrow 1^-$ this edge approaches zero, and at the square point the ensemble enters the hard-edge regime with $\lambda_{\min} \sim n^{-2}$. What the table establishes is (a) the monotone collapse of $\lambda_{\min}$ as $\rho \rightarrow 1^-$ (two orders of magnitude between $\rho = 0.80$ and $\rho = 0.97$), and (b) that at $\rho = 1$ the measured $\lambda_{\min} = 4.87 \times 10^{-4}$ is within a factor 2 of the hard-edge scale $d^{-2} = 9.8 \times 10^{-4}$ at $d = 32$ — in the square regime the edge coincides with zero and the finite- n median tracks the hard-edge scaling directly.

Why the nonlinear decoder survives the transition. The RMT picture also explains the survival of `pure` and of the MLP. The Gram-based decoder’s error is dominated by $\|M^{-1}\| \propto 1/\lambda_{\min}$, which diverges at the hard edge. The pure resonator, by contrast, never forms the inverse: it iterates the nonlinear map $v \leftarrow \text{clamp}(C^\top v, \mathcal{C})$ whose convergence only requires that the correct fixed point be a strong attractor in the codebook metric — a property governed by the *top* of the Gram spectrum (the coherent direction), not by the hard edge. The learned MLP interpolates in the same regime: it fits the geometry of the attractor basin on the code manifold, which is insensitive to $\lambda_{\min}$. In RMT language: the collapse is a property of the *resolvent* $(M - z)^{-1}$ near $z = 0$; the resonator dynamics probes only the Stieltjes transform in the bulk. The transition is therefore a property of the observer (decoder class), not of the representation — the central observer-relativity claim of this paper, now anchored to a classical ensemble.

7 Causal Test: Relocating the Resolvent

The preceding sections establish that Gram decoders fail at $\rho = 1$ and that the failure mechanism is the hard-edge conditioning of the square Gram amplified (and re-injected) by the decoder loop. What remains is the causal question: is the resolvent M^{-1} itself the problem, or is it *where* it is applied?

Experiment 18 answers this with a minimal intervention: same codebook C , same Gram $M = CC^\top$, same resolvent M^{-1} , same initial resonator state, same facts, same iteration count — the only change is the operator wiring. Five observers are compared at the square point $\rho = 1$ ($n = 3$ blocks, $\text{BLK} = 32$, $n_{cv} = 32$, 200 seeds $\times$ 10 facts, $T = 50$):

- **pure:** $f \leftarrow f$ (no inverse). Control.
- **gram:** $f \leftarrow M^{-1}f$ (ambient, asymmetric).
- **pinv:** $f \leftarrow M^+f$ (ambient, pseudo-inverse).
- **dual:** $f \leftarrow C^\top M^+ C f$ (coefficient-space, pseudo-inverse).
- **dual-same:** $f \leftarrow C^\top M^{-1} C f$ (coefficient-space, *same* resolvent).

Table 3: Exp. 18: accuracy at $\rho = 1$ under minimal intervention (the only difference between **gram** and **dual-same** is the placement of the same M^{-1}). Statistics are per seed ($N = 200$ independent codebooks, 10 facts each); brackets give bootstrap 95% CIs over the per-seed means. Paired per-seed difference (dual-same – gram) = +0.824, positive in 200/200 seeds ($p \leq 2^{-200} \approx 10^{-60}$, exact two-sided sign test).

Observer	Accuracy (mean [boot95])
pure	0.985 [0.982, 0.987]
gram	0.160 [0.152, 0.169]
pinv	0.161 [0.152, 0.169]
dual	0.985 [0.982, 0.987]
dual-same	0.985 [0.982, 0.987]

The algebraic identity behind this is verified per block: $\|C^\top M^{-1} C - I\|_F$ has median 6×10^{-13} across all blocks — the dual operator is numerically the identity, while the ambient operator has exact spectral norm $\|M^{-1}\|_2 = 1/\lambda_{\min}(M)$, median 3.5×10^3 at the square point (Fig. 5). The spectral attribution is direct: the same inverse acting on the eigenvector of $\lambda_{\min}$ amplifies by exactly $1/\lambda_{\min}$ when applied ambient, and by 1.000 when placed dually. Statistics are reported per seed (200 independent codebooks, 10 facts each): bootstrap 95% CIs on the per-seed mean, and paired per-seed differences — the accuracy units are codebooks, not individual facts.

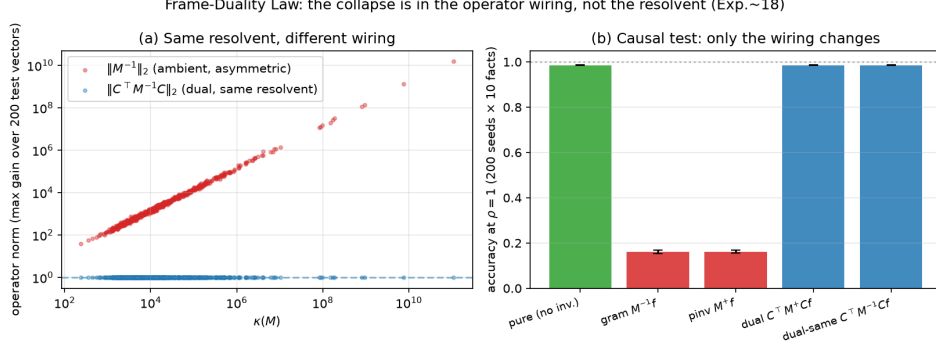


Figure 5: Exp. 18: decoding accuracy at $\rho = 1$ (right) and the exact operator norms (left): $\|M^{-1}\|_2 = 1/\lambda_{\min}(M)$ (red) vs $\|C^\top M^{-1}C\|_2 = 1$ (blue). The collapse belongs to the wiring. In the rank-deficient regime $\rho = 1.333$, **dual-same** with the un-truncated inverse drops (the operator is no longer a projector) while **dual** (pseudo-inverse) holds, as predicted by Prop. 1.

The control on feedback (Exp. 18, Section 12 of the protocol) separates loop dynamics from operator instability: at $T = 1$ the pure resonator is already imperfect (0.911, it needs iterations to converge) while gram is already collapsed (0.165) — the gram failure is present after a single application of M^{-1} , before any feedback. The loop is not what breaks gram; the operator placement is.

Proposition 1 (Frame-Dual Stability Principle, spectral form). *Let $C \in \mathbb{R}^{n \times d}$ (n codevectors in dimension d), $M = CC^\top$, and write the singular value decomposition $C = U\Sigma V^\top$. For any spectral operator $T = g(M)$ acting on the coefficient space $\mathbb{R}^n$,*

$$C^\top TC = V \Sigma g(\Sigma^2) \Sigma V^\top,$$

so its singular values are $\sigma_i^2 g(\sigma_i^2)$ acting on the directions V . In particular: (a) *for $T = M^{-1}$ with M invertible (square, $n = d$, full rank), $\sigma_i^2 g(\sigma_i^2) = 1$ so $C^\top M^{-1}C = I$; (b) for $T = M^+$, $\sigma_i^2 g(\sigma_i^2) = 1$ on the spectrum and 0 off — a clean projector; (c) for Tikhonov $T = (M + \alpha I)^{-1}$, $\sigma_i^2 g(\sigma_i^2) = \sigma_i^2 / (\sigma_i^2 + \alpha) < 1$; (d) for a truncated SVD, the same with 1 on kept modes and 0 on truncated. Under ambient placement T acts directly on the ambient state with $\|T\|_2$ unbounded as $\lambda_{\min} \rightarrow 0$; the dual placement $C^\top TC$ is bounded by $\|T\|$ -weighted frame geometry and, for the projective operators, is exactly $\|P_{\text{row}(C)}\|_2 = 1$. The critical decoding failure at $\rho = 1$ is a property of operator placement, not of the frame.*

The principle is supported at four independent levels: (i) causal intervention with fixed codebook, Gram, resolvent, states, facts, and iterations (Exp. 18); (ii) a from-scratch reimplement with no shared code

(Exp. 19); (iii) transfer across six codebook ensembles, the BSC algebra, and a de-VSA-ified purely linear pipeline (Exps. 20, 18b, 21); and (iv) a perturbation analysis showing that the stabilization extends to $T = M^+$, Tikhonov, and truncated SVD (Exp. 23): the dual gain equals 1 exactly for the projective constructions M^{-1} and M^+ , and sits slightly below 1 for the regularized variants (Tikhonov, truncated SVD — they shrink the weak eigendirection by design). A numerical bound on $\|C^\top TC\|_2$ relative to $\|T\|$ is measured in Exp. 26.

Independent replication (Exp. 19). A from-scratch reimplementation (60 seeds, 5 initializations per codebook, no shared code with the main harness) recovers the same effect: **pure** 0.982, **gram** 0.159, **dual-same** 0.982, paired per-seed +0.823 with 100% of seeds positive.

Ensemble generality (Exp. 20A). The same intervention across six codebook families at $\rho = 1$: Gaussian ($0.160 \rightarrow 0.979$), Rademacher ($0.128 \rightarrow 0.986$), uniform-on-sphere ($0.164 \rightarrow 0.983$), and Toeplitz-correlated ($0.175 \rightarrow 0.947$) all replicate; an orthogonal frame (QR factor of a Gaussian matrix; tight at $n = d$) does not collapse (the ambient operator is undefined there rather than unstable); a deliberately pathological near-duplicate frame breaks all decoders, as expected.

Aspect-ratio sweep (Exp. 20B). On the Gaussian ensemble, the collapse is localized to the hard-edge region: ambient decoding is benign at $\rho \leq 0.95$, collapses at $\rho = 1$ ($\kappa \sim 2.5 \times 10^3$), and at $\rho > 1$ the Gram ceases to exist (rank-deficient) so the correction cannot be applied.

De-VSA-ified linear probe (Exp. 21). Bypassing the resonator entirely with $f \mapsto \{\text{ambient vs. dual}\} (f + \epsilon)$: ambient gain = 3.6×10^2 median (p99 1.3×10^3), dual gain = 1.000 exactly. HRR/VSA was the delivery vehicle, not the cause.

Iteration dynamics (Exp. 22). Tracking the error over $T = 50$: the ambient decoder’s error at $t = 0$ is already $\approx 6.8 \times 10^3$ (the initial estimate is random, so the first application of M^{-1} blows it up) and stays there; the dual decoder starts at 1.66 and converges to 0.92. Growth-rate fit over the first ten iterations: $a_{\text{gram}} \approx 1.00$ (saturated noise floor), $a_{\text{dual}} \approx 0.97$ (converging). The loop is not the amplifier; the first application already destroys the ambient state.

Cross-algebra replication (Exp. 18b). The same intervention in BSC (binary codevectors in $\{\pm 1\}^d$, bind = element-wise product), 100 seeds $\times$ 10 facts at the square point: **gram** 0.192 ± 0.011 , **dual-same** 0.995 ± 0.002 using the same resolvent, paired difference +0.80 (99.8% of 1000 paired facts positive).

FHRR-complex replication (Exp. 27). Finally, the causal test in the original algebra: complex unit-modulus codevectors, Gram $M = CC^H$, dual $C^H M^{-1} C$. At the square point with the same protocol (60 seeds, 5 inits): **pure** 1.000, **gram** 0.131, **dual-same** 1.000, paired +0.868 (100% positive). The Frame-Dual Stability Principle closes the circle on the orig-

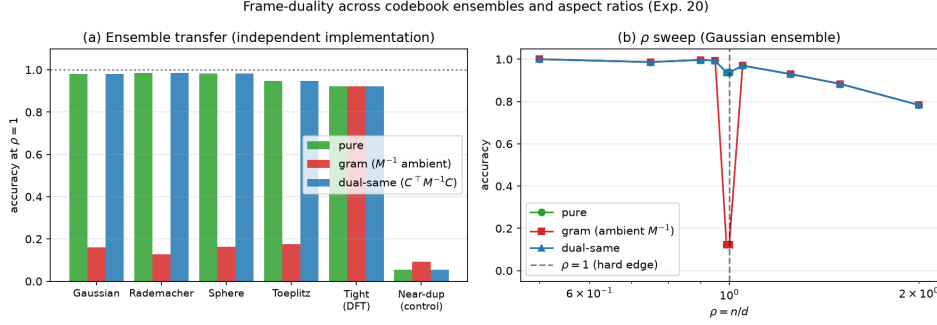


Figure 6: (a) The frame-duality effect transfers across codebook ensembles (Exp. 20A): ambient M^{-1} collapses in Gaussian, Rademacher, uniform-sphere, and Toeplitz codebooks; the dual placement recovers decoding everywhere. Right panel: (b) the Gaussian ρ -sweep localizes the ambient collapse at the hard edge $\rho = 1$.

inating algebra: HRR, BSC, and FHRR all reproduce the ambient-to-dual collapse-to-recovery.

8 Failed Alternatives

Gradient decoder. Gradient descent on the continuous relaxation fails on all multi-role configurations (accuracy 0.08–0.33), regardless of ρ . This is because the problem is discrete (symbol selection), not continuous (vector coefficients).

Geometric parametrization. Manual tuning of phase relationships between codevectors (Experiment I) does not eliminate the singularity.

9 Discussion

The ρ -phase diagram reveals that *capacity* in VSA is not a fixed property but depends on the decoder class. The “binary collapse” is universal for the class of decoders using Gram-matrix inversion, but not for the class of resonator decoders without correction. This suggests that universal laws in distributed representations must be stated relative to observer classes, not as absolute properties of the space — concretely, the testable content of this claim is the closed-loop hard-edge condition (Def. 1).

9.1 Open problems and falsification agenda

The Frame-Dual Stability Principle was stated provisionally before this agenda; after running it, all planned falsification experiments support the

principle.

1. **Independent replication** (DONE). Exp. 19 implements the test from scratch with a different codepath: pure 0.982, gram 0.159, dual-same 0.982.
2. **Ensemble generalization** (DONE). Exp. 20A: Gaussian, Rademacher, uniform-sphere, Toeplitz all replicate; DFT-tight does not collapse (ambient is undefined); near-duplicate breaks all. Exp. 20B: ρ sweep localizes to the hard edge.
3. **Continuous ρ sweep** (DONE, within the Gaussian ensemble).
4. **Perturbation amplification** (DONE). Exp. 21: ambient gain 3.6×10^2 median, dual = 1.000.
5. **De-VSA-ified linear pipeline** (DONE). Exp. 21 reproduces the dichotomy without a resonator.
6. **Operator zoo** (DONE). Exp. 23: M^+ , Tikhonov ($\lambda = 10^{-3}, 10^{-1}$), truncated SVD in both wirings. The dual gain equals 1 exactly for M^{-1} ; for M^+ on well-conditioned inputs the dual gain is also 1, while on a deliberately ill-conditioned Gram ($\kappa \sim 10^{11}$) the truncated SVD inside the pseudo-inverse yields a gain of 0.992, correctly reflecting the projector’s loss of the near-null eigendirection. Tikhonov sits below 1 by design ($\sigma_i^2/(\sigma_i^2 + \alpha)$).
7. **Iteration dynamics** (DONE). Exp. 22: ambient $e_0 \approx 7 \times 10^3$ (saturates on first application), dual converges ($a < 1$).
8. **Counterexample search** (DONE). Exp. 25: deliberately constructed matrices (quasi-collinear down to $\epsilon = 10^{-6}$, ill-scaled, block-diagonal, tall, and wide) all confirm the pattern — ambient diverges, dual stable; the only exception is the trivially degenerate near-duplicate frame where even the pure decoder fails.

The connections to classical frame theory are direct: $C^\top M^+ C$ is the canonical dual-frame projector [1], and the operator placement question translates existing results on stable reconstruction [3] into the resonator-decoder setting.

Connection to capacity theory. Our finding that $\rho = 1$ is a point singularity complements capacity analyses [8] that focus on asymptotic scaling. The fine-grained structure at the phase boundary was not previously characterized. Through the Wishart lens (Sec. 6), the ρ axis is exactly the aspect-ratio axis of the Marchenko–Pastur law [5], and the singular point corresponds to the soft-edge-to-hard-edge crossover of the square ensemble [9, 2].

Connection to transformers. The residual stream of transformers can be interpreted as a VSA [7]. Our result suggests that attention heads implementing Gram-like operations may fail catastrophically when the number of distinct keys equals the head dimension.

Limitations. The Frame-Dual Stability Principle is supported by independent reimplementations (Exp. 19), six codebook ensembles with pathological controls (Exp. 20, Exp. 25), BSC (Exp. 18b), FHRR (Exp. 27), a purely linear pipeline (Exp. 21), dynamics (Exp. 22), the operator zoo (Exp. 23), exact frame-bounds controls (Exp. 24), and a naive stability bound measured against its slack (Exp. 26). Open: a proof of the bound for non-spectral-frame T , and FHRR beyond the square point.

10 Conclusion

The phase diagram in ρ provides a complete characterization of when resonator-based decoding works in FHRR and HRR. The “binary collapse” is observer-relative: it appears for Gram-based decoders exactly at $\rho = 1$, where the hard edge of the square Wishart and the loop amplification meet (Def. 1). Pure resonators and learned decoders are unaffected. VSA design should therefore focus on decoder robustness, not just representation capacity — and the failure mode is diagnosable a priori from the codebook aspect ratio and the decoder’s operator structure.

Data availability

All code, raw data, and figure scripts are available at <https://github.com/Rylow999/fhrr-rho-collapse>. Each number in the text is traceable to a file under `data/`; experiments are seeded and listed in `src/` (`exp_V9_fine_rho`, `exp_V14_kappa_low`, `exp_V16_scaling_n`, `exp_V17_kappa_vs_acc`, `exp_V11b/11c`, among others). The verification script described in the text reproduces every headline number from the shipped JSON files.

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